

Time Harmonic E & H Fields

SAY WE HAVE A TIME HARMONIC
 $\bar{E}$ FIELD : $\bar{E}(x, y, z; t)$

$$\bar{E}(x, y, z; t) = \text{Re} \left[\underbrace{\tilde{E}(x, y, z)}_{\text{PHASOR}} e^{j\omega t} \right]$$

$$\bar{H}(x, y, z; t) = \text{Re} \left[\underbrace{\tilde{H}(x, y, z)} e^{j\omega t} \right]$$

$$k = \frac{\omega}{u_p}$$

is called wavenumber or phase constant for lossless media
In general, the phase constant is denoted by β

Maxwell's Equations in Phasor form

$$\vec{\nabla} \cdot \vec{E} = \rho / \epsilon_0$$

$$\vec{\nabla} \cdot \tilde{E} = \rho_v / \epsilon$$

Gauss's law for electric fields

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\vec{\nabla} \cdot \tilde{B} = 0$$

Gauss's law for magnetic fields

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{aligned} \vec{\nabla} \times \tilde{E} &= -j\omega \tilde{B} \\ &= -j\omega \mu \tilde{H} \\ [\tilde{B} &= \mu \tilde{H}] \end{aligned}$$

Faraday's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\begin{aligned} \vec{\nabla} \times \tilde{H} &= \tilde{J} + j\omega \tilde{D} \\ &= \tilde{J} + j\omega \epsilon \tilde{E} \end{aligned}$$



Ampere-Maxwell law

Maxwell's Equations

For sinusoidal time variations:

$$\begin{aligned}\nabla \cdot \tilde{\mathbf{E}} &= \tilde{\rho}_v / \varepsilon, \\ \nabla \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}}, \\ \nabla \cdot \tilde{\mathbf{H}} &= 0, \\ \nabla \times \tilde{\mathbf{H}} &= \tilde{\mathbf{J}} + j\omega\varepsilon\tilde{\mathbf{E}}.\end{aligned}$$

$$\begin{aligned}\nabla \times \tilde{\mathbf{H}} &= \tilde{\mathbf{J}} + j\omega\varepsilon\tilde{\mathbf{E}} \\ &= (\sigma + j\omega\varepsilon)\tilde{\mathbf{E}} = j\omega\left(\varepsilon - j\frac{\sigma}{\omega}\right)\tilde{\mathbf{E}}.\end{aligned}$$

By defining the *complex permittivity* ε_c as

$$\varepsilon_c = \varepsilon - j\frac{\sigma}{\omega}, \quad (7.4)$$

Eq. (7.3) can be rewritten as

$$\nabla \times \tilde{\mathbf{H}} = j\omega\varepsilon_c\tilde{\mathbf{E}}.$$

In a source-free medium

$$\tilde{\rho}_v = 0$$

Consequently, Maxwell's equations become:

$$\begin{aligned}\nabla \cdot \tilde{\mathbf{E}} &= 0, \\ \nabla \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}}, \\ \nabla \cdot \tilde{\mathbf{H}} &= 0, \\ \nabla \times \tilde{\mathbf{H}} &= j\omega\varepsilon_c\tilde{\mathbf{E}}.\end{aligned}$$

We will use these to derive the wave equation for EM waves.

Complex Permittivity

$$\varepsilon_c = \varepsilon - j \frac{\sigma}{\omega} = \varepsilon' - j\varepsilon'', \quad (7.7)$$

with

$$\varepsilon' = \varepsilon, \quad (7.8a)$$

$$\varepsilon'' = \frac{\sigma}{\omega}. \quad (7.8b)$$

For a lossless medium with $\sigma = 0$, it follows that $\varepsilon'' = 0$ and $\varepsilon_c = \varepsilon' = \varepsilon$.

A lossless medium is typically associated with dielectric materials rather than conductors. In a lossless medium, there is zero electrical conductivity ($\sigma = 0$), meaning that electrical charges within the material do not move in response to an applied electric field. This lack of motion prevents the dissipation of electrical energy as heat, which is a characteristic of dielectric materials.

Wave Equations

$$\nabla \cdot \tilde{\mathbf{E}} = 0, \quad (7.6a)$$

$$\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}, \quad (7.6b)$$

$$\nabla \cdot \tilde{\mathbf{H}} = 0, \quad (7.6c)$$

$$\nabla \times \tilde{\mathbf{H}} = j\omega\epsilon_c\tilde{\mathbf{E}}. \quad (7.6d)$$

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(\nabla \times \tilde{\mathbf{H}}). \quad (7.9)$$

Upon substituting Eq. (7.6d) into Eq. (7.9) we obtain

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = -j\omega\mu(j\omega\epsilon_c\tilde{\mathbf{E}}) = \omega^2\mu\epsilon_c\tilde{\mathbf{E}}. \quad (7.10)$$

From Eq. (3.113), we know that the curl of the curl of $\tilde{\mathbf{E}}$ is

$$\nabla \times (\nabla \times \tilde{\mathbf{E}}) = \nabla(\nabla \cdot \tilde{\mathbf{E}}) - \nabla^2\tilde{\mathbf{E}}, \quad (7.11)$$

where $\nabla^2\tilde{\mathbf{E}}$ is the Laplacian of $\tilde{\mathbf{E}}$, which in Cartesian coordinates is given by

$$\nabla^2\tilde{\mathbf{E}} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \tilde{\mathbf{E}}. \quad (7.12)$$

In view of Eq. (7.6a), the use of Eq. (7.11) in Eq. (7.10) gives

$$\nabla^2\tilde{\mathbf{E}} + \omega^2\mu\epsilon_c\tilde{\mathbf{E}} = 0, \quad (7.13)$$

which is known as the *homogeneous wave equation for $\tilde{\mathbf{E}}$* . By defining the *propagation constant γ* as

$$\gamma^2 = -\omega^2\mu\epsilon_c, \quad (7.14)$$

Eq. (7.13) can be written as

$$\nabla^2\tilde{\mathbf{E}} - \gamma^2\tilde{\mathbf{E}} = 0. \quad (7.15)$$

To derive Eq. (7.15), we took the curl of both sides of Eq. (7.6b) and then we used Eq. (7.6d) to eliminate $\tilde{\mathbf{H}}$ and obtain an equation in $\tilde{\mathbf{E}}$ only. If we reverse the process, that is, if we start by taking the curl of both sides of Eq. (7.6d) and then use Eq. (7.6b) to eliminate $\tilde{\mathbf{E}}$, we obtain a wave equation for $\tilde{\mathbf{H}}$:

$$\nabla^2\tilde{\mathbf{H}} - \gamma^2\tilde{\mathbf{H}} = 0. \quad (7.16)$$

Since the wave equations for $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ are of the same form, so are their solutions.

Lossless Media

If the medium is **nonconducting** ($\sigma = 0$), the wave does not suffer any attenuation as it travels and hence the medium is said to be **lossless**.

$$\gamma^2 = -\omega^2 \mu \epsilon. \quad (7.17)$$

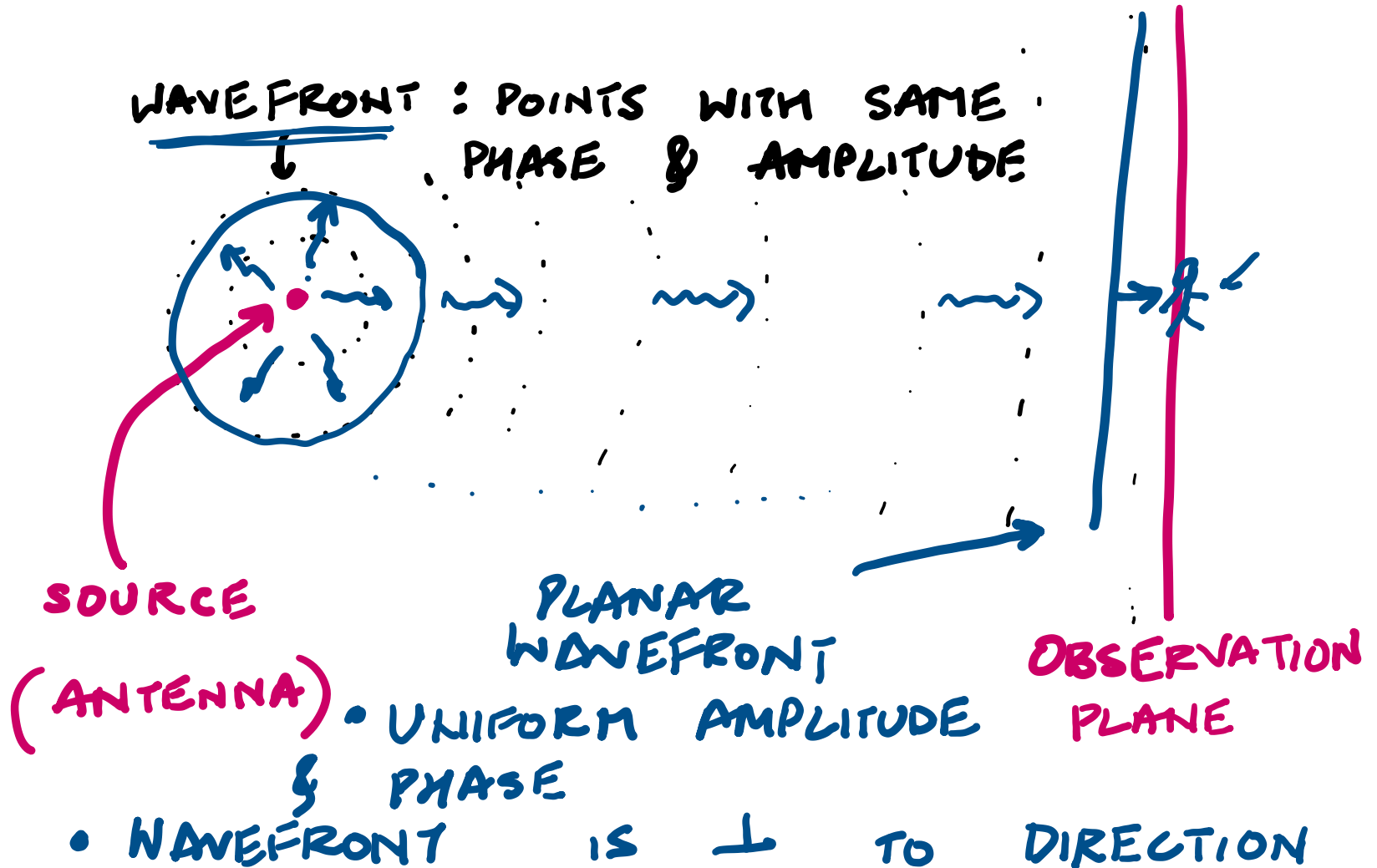
For lossless media, it is customary to define the *wavenumber* k as

$$k = \omega \sqrt{\mu \epsilon}. \quad (7.18)$$

In view of Eq. (7.17), $\gamma^2 = -k^2$ and Eq. (7.15) becomes

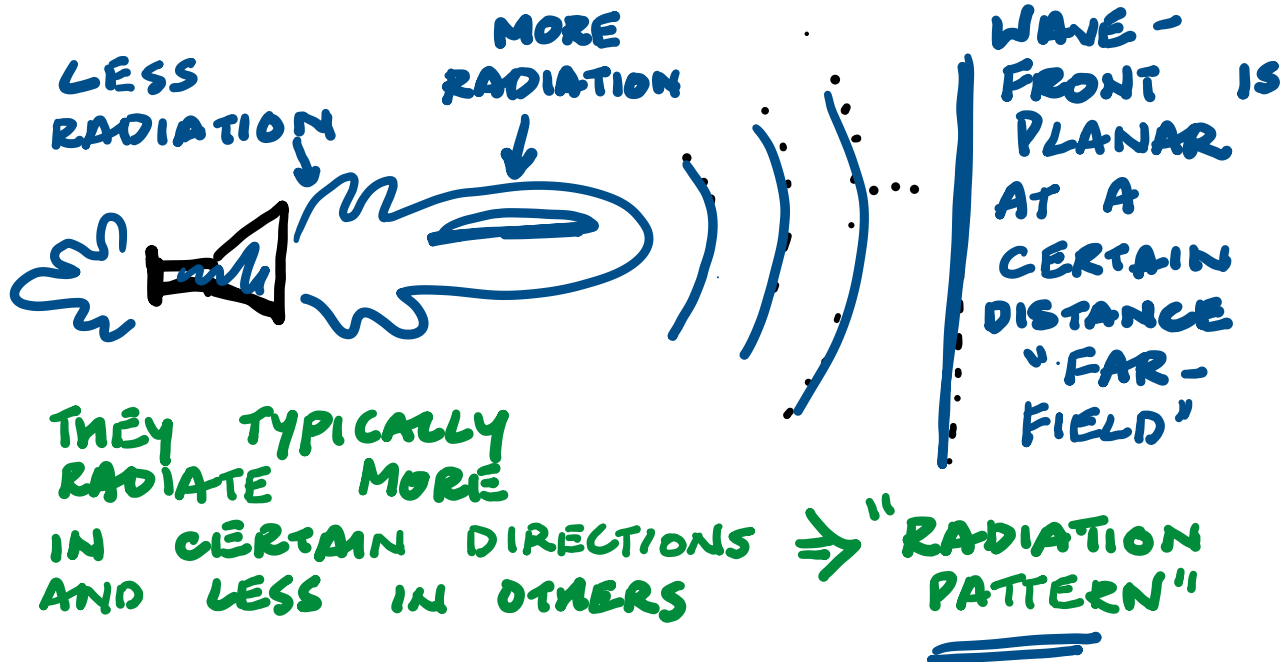
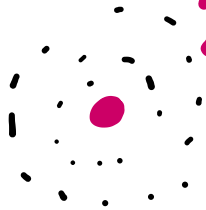
$$\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0. \quad (7.19)$$

What is a plane wave?

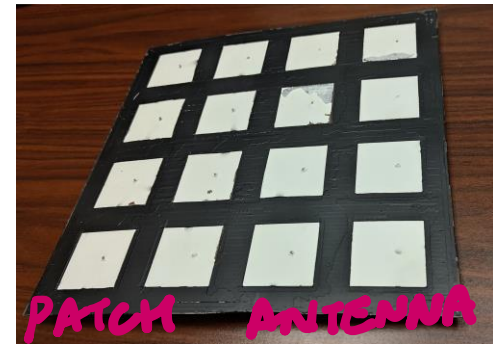


What is a plane wave?

SOURCES / ANTENNAS IN REAL LIFE :
TYPICALLY ARE NOT
POINT SOURCES THAT
RADIATE SPHERICALLY



HORN
ANTENNA

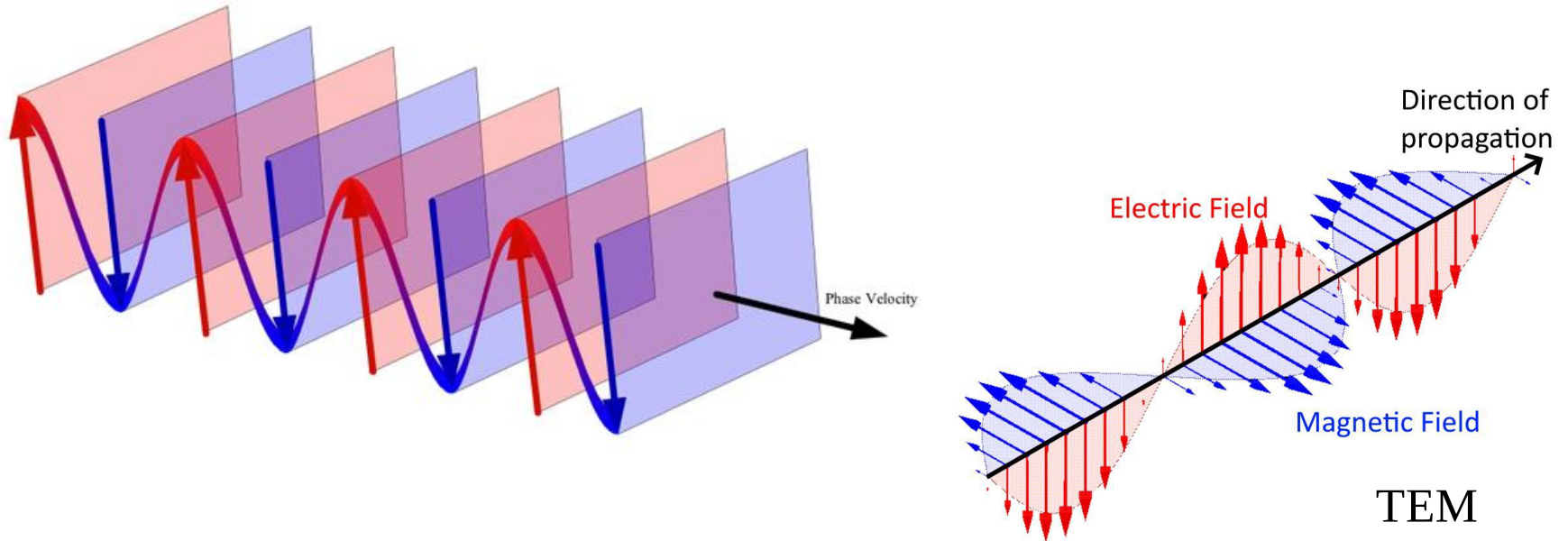


PATCH ANTENNA
ARRAY



YAGI-UDA
ANTENNA

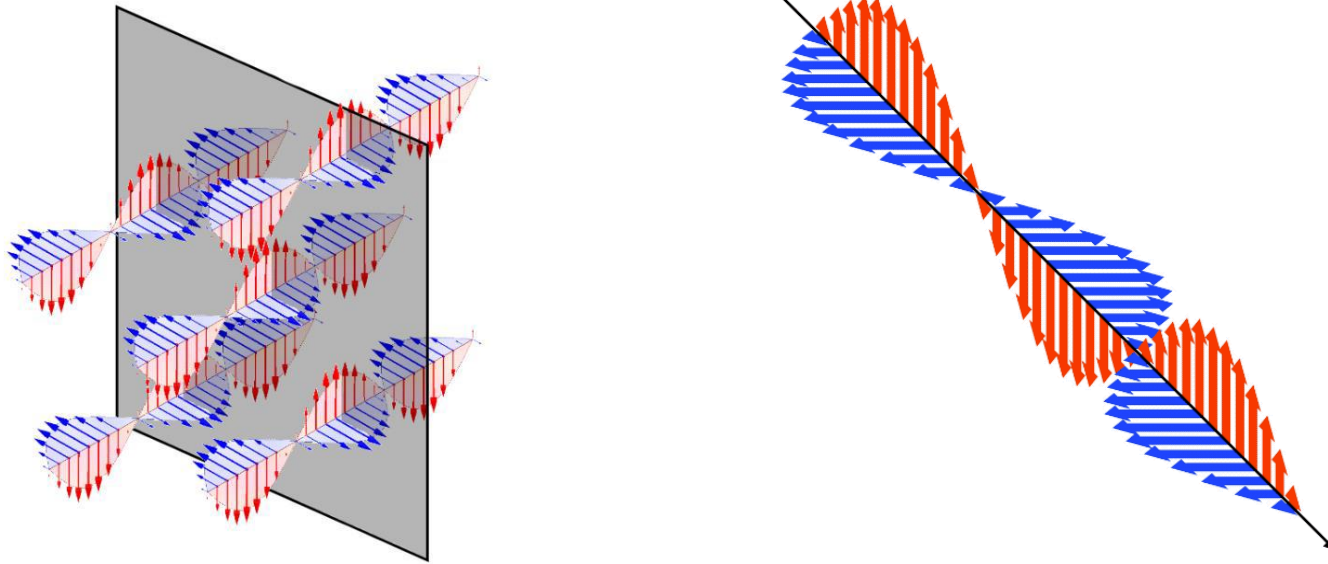
What is a plane wave?



Plane waves:

1. Are Transverse Electromagnetic (TEM) Waves
 $\mathbf{E}$ and $\mathbf{H}$ are perpendicular to one another and to the direction of propagation $\mathbf{k}$
2. Are invariant over a plane normal to the direction of propagation

What is a plane wave?



Plane wave's magnitude, phase, and orientation must depend only on the variable(s) associated with the direction of propagation

What is a plane wave?

Plane wave's magnitude, phase, and orientation must depend only on the variable(s) associated with the direction of propagation

Consider: A TEM wave has the following electric field:

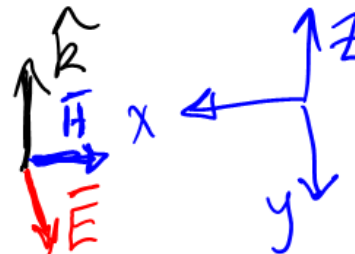
$$\tilde{\vec{E}} = E_o e^{-j2z} \hat{y} \quad (\text{phasor domain})$$

$$\vec{E} = E_o \cos(\omega t - 2z) \hat{y} \quad (\text{time domain})$$

from which we may observe that:

- The wave is propagating in the +z direction
- The electric field is oriented along the y-axis
- Since the wave is TEM, the magnetic field will be oriented along the x-axis
- E_o is the magnitude of the electric field vector.

Note: If E_o does not vary as a function of x or y, this equation represents a plane wave.



Plane Wave

$$\vec{E} = 3e^{-j2z} \hat{y}$$

Not a Plane Wave

$$\vec{E} = \frac{3}{x} e^{-j2z} \hat{y}$$

What is a plane wave?

- Plane waves are mathematically convenient constructs we can use to derive general principles about electromagnetic waves.
- Plane waves are simplified idealizations of electromagnetic waves (no real wave is perfectly constant over a plane).
- Waves may be approximated as plane waves with great accuracy (e.g. when far from the source).
- Electromagnetic wave properties are often derived with respect to plane waves, because of the math simplicity. However, the principles derived from plane wave analysis are often generalizable to all EM waves.

Uniform Plane Wave

Recall: $\nabla^2 \tilde{\mathbf{E}} + k^2 \tilde{\mathbf{E}} = 0.$ (7.19)

For an electric field phasor decomposed in its Cartesian components as

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z, \quad (7.20)$$

substitution of Eq. (7.12) into Eq. (7.19) gives

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) + k^2(\hat{\mathbf{x}}\tilde{E}_x + \hat{\mathbf{y}}\tilde{E}_y + \hat{\mathbf{z}}\tilde{E}_z) = 0. \quad (7.21)$$

To satisfy Eq. (7.21), each vector component on the left-hand side of the equation must vanish. Hence,

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2 \right) \tilde{E}_x = 0, \quad (7.22)$$

and similar expressions apply to $\tilde{E}_y$ and $\tilde{E}_z$.

Cont.

Uniform Plane Wave

A uniform plane wave is characterized by electric and magnetic fields that have uniform properties at all points across an infinite plane.

If this happens to be the x - y plane, then $\mathbf{E}$ and $\mathbf{H}$ do not vary with x and y . Hence, $\partial \tilde{E}_x / \partial x = 0$ and $\partial \tilde{E}_x / \partial y = 0$, and Eq. (7.22) reduces to

$$\frac{d^2 \tilde{E}_x}{dz^2} + k^2 \tilde{E}_x = 0. \quad (7.23)$$

General Form of the Solution:

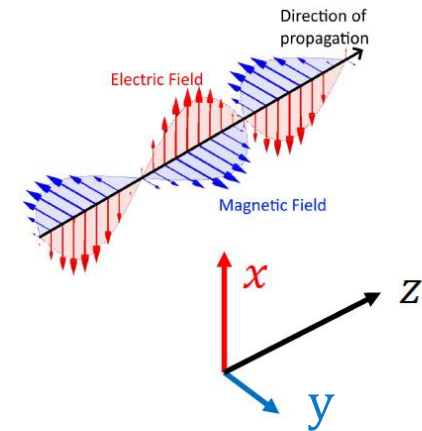
$$\tilde{E}_x(z) = \tilde{E}_x^+(z) + \tilde{E}_x^-(z) = E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}$$

For a wave travelling along +z only:

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz}$$

Application of $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$ yields:

$$\tilde{H}_y(z) = \frac{k}{\omega\mu} E_{x0}^+ e^{-jkz} = H_{y0}^+ e^{-jkz}$$



Summary: This is a plane wave with

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} \tilde{E}_x^+(z) = \hat{\mathbf{x}} E_{x0}^+ e^{-jkz},$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}} \frac{E_{x0}^+}{\eta} e^{-jkz}.$$

with

$$\eta = \frac{\omega\mu}{k} = \frac{\omega\mu}{\omega\sqrt{\mu\epsilon}} = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega)$$

Intrinsic impedance of a lossless media is defined in analogy by characteristic impedance of a transmission line.

Intrinsic Impedance of Plane Waves

- A plane wave has a fixed ratio of electric field magnitude to magnetic field magnitude

$$\eta = \frac{E_x^+}{H_y^+} = -\frac{E_x^-}{H_y^-} = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\epsilon}}$$

Or, in words,

$$\text{intrinsic impedance} = \frac{\text{Forward E}}{\text{Forward H}} = -\frac{\text{Backward E}}{\text{Backward H}} = \sqrt{\frac{\mu}{\epsilon}}$$

Note: in free space,

$$\eta_o = \sqrt{\frac{\mu_o}{\epsilon_o}} = 377\Omega$$

Summary from previous slide:

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\tilde{E}_x^+(z) = \hat{\mathbf{x}}E_{x0}^+e^{-jkz},$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\frac{\tilde{E}_x^+(z)}{\eta} = \hat{\mathbf{y}}\frac{E_{x0}^+}{\eta}e^{-jkz}.$$

Time-Domain Solution

In the general case, E_{x0}^+ is a complex quantity with magnitude $|E_{x0}^+|$ and phase angle ϕ^+ . That is,

$$E_{x0}^+ = |E_{x0}^+|e^{j\phi^+}. \quad (7.33)$$

The instantaneous electric and magnetic fields therefore are

$$\begin{aligned} \mathbf{E}(z, t) &= \Re \left[\tilde{\mathbf{E}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{x}}|E_{x0}^+| \cos(\omega t - kz + \phi^+) \quad (\text{V/m}), \end{aligned} \quad (7.34a)$$

and

$$\begin{aligned} \mathbf{H}(z, t) &= \Re \left[\tilde{\mathbf{H}}(z) e^{j\omega t} \right] \\ &= \hat{\mathbf{y}}\frac{|E_{x0}^+|}{\eta} \cos(\omega t - kz + \phi^+) \quad (\text{A/m}). \end{aligned} \quad (7.34b)$$

Wave's Phase Velocity

$$u_p = \frac{\omega}{k} = \frac{\omega}{\omega\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{\mu\varepsilon}} \quad (\text{m/s}), \quad (7.35)$$

and its wavelength is

$$\lambda = \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m}). \quad (7.36)$$

In vacuum, $\varepsilon = \varepsilon_0$ and $\mu = \mu_0$, and the phase velocity u_p and the intrinsic impedance η given by Eq. (7.31) are

$$u_p = c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 3 \times 10^8 \quad (\text{m/s}), \quad (7.37)$$

$$\eta = \eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 377 \, (\Omega) \approx 120\pi \, (\Omega), \quad (7.38)$$

Example 7-1: EM Plane Wave in Air

This example is analogous to the “Sound Wave in Water” problem given by Example 1-1.

The electric field of a 1-MHz plane wave traveling in the $+z$ -direction in air points along the x -direction. If this field reaches a peak value of 1.2π (mV/m) at $t = 0$ and $z = 50$ m, obtain expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$, and then plot them as a function of z at $t = 0$.

Solution: At $f = 1$ MHz, the wavelength in air is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m},$$

and the corresponding wavenumber is $k = (2\pi/300)$ (rad/m). The general expression for an x -directed electric field traveling in the $+z$ -direction is given by Eq. (7.34a) as

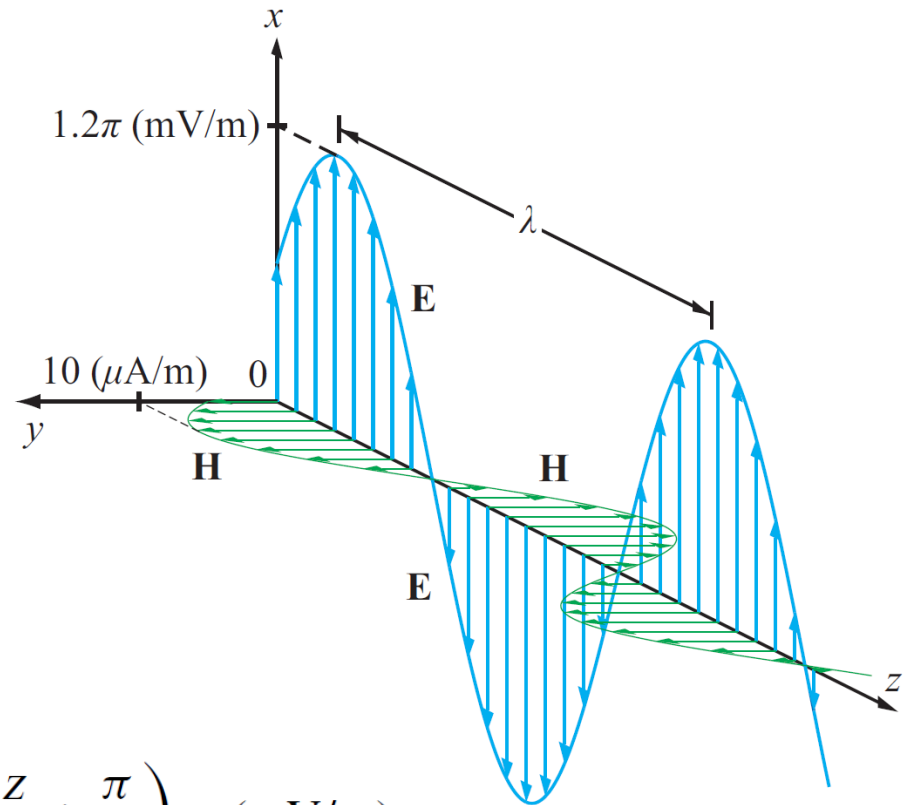
$$\begin{aligned}\mathbf{E}(z, t) &= \hat{\mathbf{x}} |E_{x0}^+| \cos(\omega t - kz + \phi^+) \\ &= \hat{\mathbf{x}} 1.2\pi \cos\left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \phi^+\right) \text{ (mV/m)}.\end{aligned}$$

The field $\mathbf{E}(z, t)$ is maximum when the argument of the cosine function equals zero or a multiple of 2π . At $t = 0$ and $z = 50$ m, this condition yields

$$-\frac{2\pi \times 50}{300} + \phi^+ = 0 \quad \text{or} \quad \phi^+ = \frac{\pi}{3}.$$

Cont.

Example 7-1 cont.



Hence,

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 1.2\pi \cos \left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3} \right) \quad (\text{mV/m}),$$

and from Eq. (7.34b) we have

$$\begin{aligned} \mathbf{H}(z, t) &= \hat{\mathbf{y}} \frac{E(z, t)}{\eta_0} \\ &= \hat{\mathbf{y}} 10 \cos \left(2\pi \times 10^6 t - \frac{2\pi z}{300} + \frac{\pi}{3} \right) \quad (\mu\text{A/m}), \end{aligned}$$

Directional Relation Between **E** and **H**

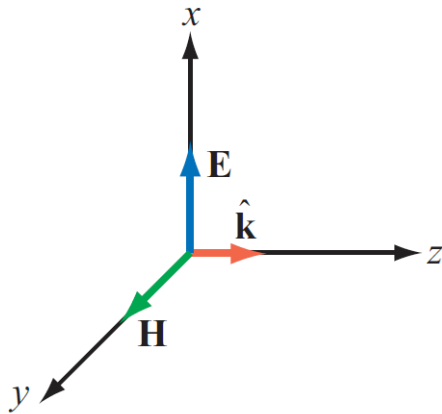


Figure 7-4: A transverse electromagnetic (TEM) wave propagating in the direction $\hat{\mathbf{k}} = \hat{\mathbf{z}}$. For all TEM waves, $\hat{\mathbf{k}}$ is parallel to $\mathbf{E} \times \mathbf{H}$.

For Any TEM Wave

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}}, \quad (7.39a)$$

$$\tilde{\mathbf{E}} = -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}. \quad (7.39b)$$

*The following right-hand rule applies: when we rotate the four fingers of the right hand from the direction of **E** toward that of **H**, the thumb points in the direction of wave travel, $\hat{\mathbf{k}}$.*

Wave decomposition

In general, a uniform plane wave traveling in the +z-direction may have both x- and y-components, in which case $\tilde{\mathbf{E}}$ is given by

$$\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x^+(z) + \hat{\mathbf{y}} \tilde{E}_y^+(z), \quad (7.43a)$$

and the associated magnetic field is

$$\tilde{\mathbf{H}} = \hat{\mathbf{x}} \tilde{H}_x^+(z) + \hat{\mathbf{y}} \tilde{H}_y^+(z). \quad (7.43b)$$

Application of Eq. (7.39a) gives

$$\tilde{\mathbf{H}} = \frac{1}{\eta} \hat{\mathbf{z}} \times \tilde{\mathbf{E}} = -\hat{\mathbf{x}} \frac{\tilde{E}_y^+(z)}{\eta} + \hat{\mathbf{y}} \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.44)$$

By equating Eq. (7.43b) to Eq. (7.44), we have

$$\tilde{H}_x^+(z) = -\frac{\tilde{E}_y^+(z)}{\eta}, \quad \tilde{H}_y^+(z) = \frac{\tilde{E}_x^+(z)}{\eta}. \quad (7.45)$$

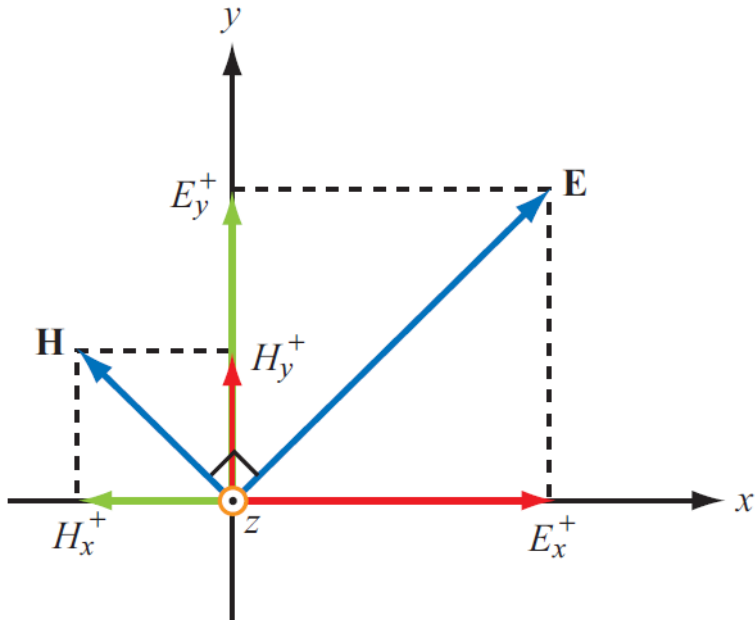


Figure 7-6: The wave ($\mathbf{E}$, $\mathbf{H}$) is equivalent to the sum of two waves, one with fields (E_x^+ , H_y^+) and another with (E_y^+ , H_x^+) with both traveling in the +z-direction.

Lossy Media

For a uniform plane wave with electric field $\tilde{\mathbf{E}} = \hat{\mathbf{x}} \tilde{E}_x(z)$ traveling along the z -direction, the wave equation given by Eq. (7.61) reduces to

$$\frac{d^2 \tilde{E}_x(z)}{dz^2} - \gamma^2 \tilde{E}_x(z) = 0. \quad (7.67)$$

with

$$\gamma^2 = -\omega^2 \mu \epsilon_c = -\omega^2 \mu (\epsilon' - j\epsilon''), \quad (7.62)$$

where $\epsilon' = \epsilon$ and $\epsilon'' = \sigma/\omega$. Since γ is complex, we express it as

$$\gamma = \alpha + j\beta, \quad (7.63)$$

where α is the medium's *attenuation constant* and β its *phase constant*. By replacing γ with $(\alpha + j\beta)$ in Eq. (7.62), we obtain

$$\begin{aligned} (\alpha + j\beta)^2 &= (\alpha^2 - \beta^2) + j2\alpha\beta \\ &= -\omega^2 \mu \epsilon' + j\omega^2 \mu \epsilon''. \end{aligned} \quad (7.64)$$

Cont.

Lossy Media

The rules of complex algebra require the real and imaginary parts on one side of an equation to equal the real and imaginary parts on the other side. Hence,

$$\alpha^2 - \beta^2 = -\omega^2 \mu \varepsilon', \quad (7.65a)$$

$$2\alpha\beta = \omega^2 \mu \varepsilon''. \quad (7.65b)$$

Solving these two equations for α and β gives

$$\alpha = \omega \left\{ \frac{\mu \varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right] \right\}^{1/2} \quad (\text{Np/m}), \quad (7.66a)$$

$$\beta = \omega \left\{ \frac{\mu \varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right] \right\}^{1/2} \quad (\text{rad/m}). \quad (7.66b)$$

Attenuation

E and H fields:

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\tilde{E}_x(z) = \hat{\mathbf{x}}E_{x0}e^{-\gamma z} = \hat{\mathbf{x}}E_{x0}e^{-\alpha z}e^{-j\beta z}. \quad (7.68)$$

The associated magnetic field $\tilde{\mathbf{H}}$ can be determined by applying Eq. (7.2b): $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$, or using Eq. (7.39a): $\tilde{\mathbf{H}} = (\hat{\mathbf{k}} \times \tilde{\mathbf{E}})/\eta_c$, where η_c is the *intrinsic impedance of the lossy medium*. Both approaches give

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\tilde{H}_y(z) = \hat{\mathbf{y}}\frac{\tilde{E}_x(z)}{\eta_c} = \hat{\mathbf{y}}\frac{E_{x0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}, \quad (7.69)$$

where

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j\frac{\epsilon''}{\epsilon'}\right)^{-1/2} \quad (\Omega). \quad (7.70)$$

Cont.

Attenuation

Magnitude of E

$$|\tilde{E}_x(z)| = |E_{x0}e^{-\alpha z}e^{-j\beta z}| = |E_{x0}|e^{-\alpha z}$$

Skin depth

$$\delta_s = \frac{1}{\alpha} \quad (\text{m}), \quad (7.72)$$

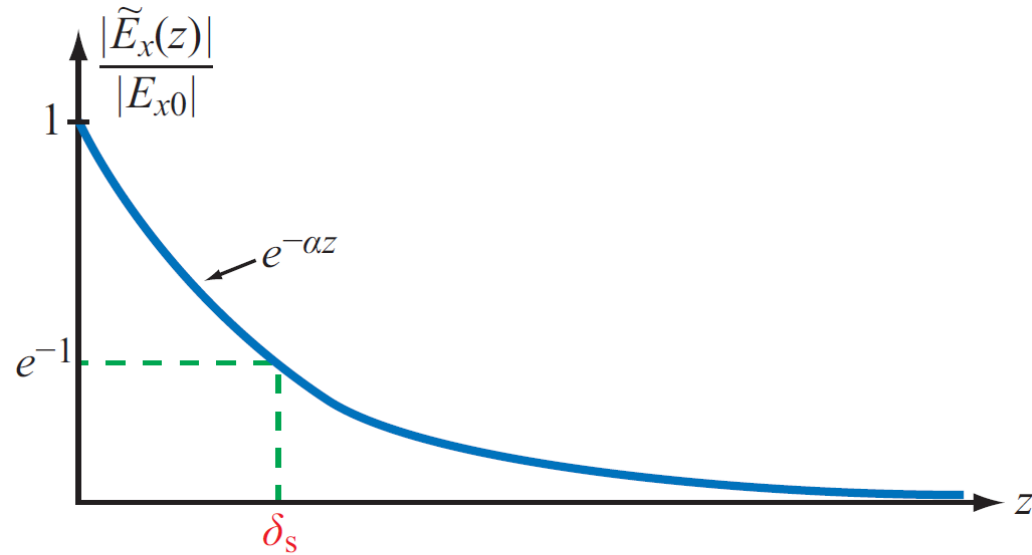


Figure 7-13: Attenuation of the magnitude of $\tilde{E}_x(z)$ with distance z . The skin depth δ_s is the value of z at which $|\tilde{E}_x(z)|/|E_{x0}| = e^{-1}$, or $z = \delta_s = 1/\alpha$.

the wave magnitude decreases by a factor of $e^{-1} \approx 0.37$ (Fig. 7-13). At depth $z = 3\delta_s$, the field magnitude is less than 5% of its initial value, and at $z = 5\delta_s$, it is less than 1%.

*This distance δ_s , called the **skin depth** of the medium, characterizes how deep an electromagnetic wave can penetrate into a conducting medium.*

Low and High Frequency Approximations

Table 7-1: Expressions for α , β , η_c , u_p , and λ for various types of media.

	Any Medium	Lossless Medium ($\sigma = 0$)	Low-loss Medium ($\varepsilon''/\varepsilon' \ll 1$)	Good Conductor ($\varepsilon''/\varepsilon' \gg 1$)	Units
$\alpha =$	$\omega \left[\frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right] \right]^{1/2}$	0	$\frac{\sigma}{2} \sqrt{\frac{\mu}{\varepsilon}}$	$\sqrt{\pi f \mu \sigma}$	(Np/m)
$\beta =$	$\omega \left[\frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu\varepsilon}$	$\omega \sqrt{\mu\varepsilon}$	$\sqrt{\pi f \mu \sigma}$	(rad/m)
$\eta_c =$	$\sqrt{\frac{\mu}{\varepsilon'}} \left(1 - j \frac{\varepsilon''}{\varepsilon'} \right)^{-1/2}$	$\sqrt{\frac{\mu}{\varepsilon}}$	$\sqrt{\frac{\mu}{\varepsilon}}$	$(1 + j) \frac{\alpha}{\sigma}$	(Ω)
$u_p =$	ω / β	$1 / \sqrt{\mu\varepsilon}$	$1 / \sqrt{\mu\varepsilon}$	$\sqrt{4\pi f / \mu \sigma}$	(m/s)
$\lambda =$	$2\pi / \beta = u_p / f$	u_p / f	u_p / f	u_p / f	(m)
Notes: $\varepsilon' = \varepsilon$; $\varepsilon'' = \sigma/\omega$; in free space, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$; in practice, a material is considered a low-loss medium if $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon < 0.01$ and a good conducting medium if $\varepsilon''/\varepsilon' > 100$.					

Example 7-4: Plane Wave in Seawater

A uniform plane wave is traveling in seawater. Assume that the x - y plane resides just below the sea surface and the wave travels in the $+z$ -direction into the water. The constitutive parameters of seawater are $\epsilon_r = 80$, $\mu_r = 1$, and $\sigma = 4$ S/m. If the magnetic field at $z = 0$ is $\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ)$ (mA/m),

- (a) obtain expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$, and
- (b) determine the depth at which the magnitude of $\mathbf{E}$ is 1% of its value at $z = 0$.

Solution: (a) Since $\mathbf{H}$ is along $\hat{\mathbf{y}}$ and the propagation direction is $\hat{\mathbf{z}}$, $\mathbf{E}$ must be along $\hat{\mathbf{x}}$. Hence, the general expressions for the phasor fields are

$$\tilde{\mathbf{E}}(z) = \hat{\mathbf{x}} E_{x0} e^{-\alpha z} e^{-j\beta z}, \quad (7.78a)$$

$$\tilde{\mathbf{H}}(z) = \hat{\mathbf{y}} \frac{E_{x0}}{\eta_c} e^{-\alpha z} e^{-j\beta z}. \quad (7.78b)$$

Example 7-4: Plane Wave in Seawater

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0} = \frac{4}{2\pi \times 10^3 \times 80 \times (10^{-9}/36\pi)} \\ = 9 \times 10^5.$$

This qualifies seawater as a good conductor at 1 kHz and allows us to use the good-conductor expressions given in Table 7-1:

$$\alpha = \sqrt{\pi f \mu \sigma} \\ = \sqrt{\pi \times 10^3 \times 4\pi \times 10^{-7} \times 4} \\ = 0.126 \quad (\text{Np/m}), \quad (7.79a)$$

$$\beta = \alpha = 0.126 \quad (\text{rad/m}), \quad (7.79b)$$

$$\eta_c = (1 + j) \frac{\alpha}{\sigma} \\ = (\sqrt{2} e^{j\pi/4}) \frac{0.126}{4} = 0.044 e^{j\pi/4} \quad (\Omega). \quad (7.79c)$$

Example 7-4: Plane Wave in Seawater

$$\begin{aligned}\mathbf{E}(z, t) &= \Re \left[\hat{\mathbf{x}} |E_{x0}| e^{j\phi_0} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right] \\ &= \hat{\mathbf{x}} |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + \phi_0) \\ &\quad (\text{V/m}),\end{aligned}\tag{7.80a}$$

$$\begin{aligned}\mathbf{H}(z, t) &= \Re \left[\hat{\mathbf{y}} \frac{|E_{x0}| e^{j\phi_0}}{0.044 e^{j\pi/4}} e^{-\alpha z} e^{-j\beta z} e^{j\omega t} \right] \\ &= \hat{\mathbf{y}} 22.5 |E_{x0}| e^{-0.126z} \cos(2\pi \times 10^3 t \\ &\quad - 0.126z + \phi_0 - 45^\circ) \quad (\text{A/m}).\end{aligned}\tag{7.80b}$$

Example 7-4: Plane Wave in Seawater

At $z = 0$,

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 22.5 |E_{x0}| \cos(2\pi \times 10^3 t + \phi_0 - 45^\circ) \quad (\text{A/m}). \quad (7.81)$$

By comparing Eq. (7.81) with the expression given in the problem statement,

$$\mathbf{H}(0, t) = \hat{\mathbf{y}} 100 \cos(2\pi \times 10^3 t + 15^\circ) \quad (\text{mA/m}),$$

we deduce that

$$22.5 |E_{x0}| = 100 \times 10^{-3}$$

or

$$|E_{x0}| = 4.44 \quad (\text{mV/m}),$$

and

$$\phi_0 - 45^\circ = 15^\circ \quad \text{or} \quad \phi_0 = 60^\circ.$$

Cont.

Example 7-4: Plane Wave in Seawater

Hence, the final expressions for $\mathbf{E}(z, t)$ and $\mathbf{H}(z, t)$ are

$$\mathbf{E}(z, t) = \hat{\mathbf{x}} 4.44 e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + 60^\circ) \quad (\text{mV/m}), \quad (7.82a)$$

$$\mathbf{H}(z, t) = \hat{\mathbf{y}} 100 e^{-0.126z} \cos(2\pi \times 10^3 t - 0.126z + 15^\circ) \quad (\text{mA/m}). \quad (7.82b)$$

(b) The depth at which the amplitude of $\mathbf{E}$ has decreased to 1% of its initial value at $z = 0$ is obtained from

$$0.01 = e^{-0.126z}$$

or

$$z = \frac{\ln(0.01)}{-0.126} = 36.55 \text{ m} \approx 37 \text{ m}$$

Power Density

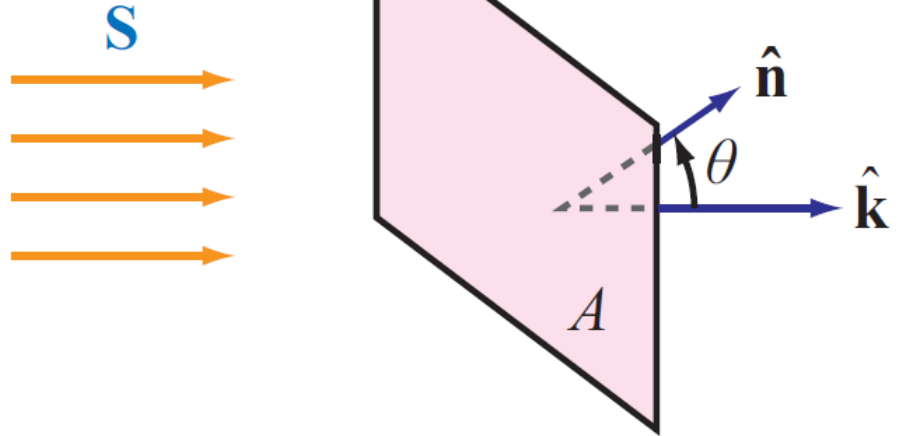
Electromagnetic waves carry **energy**. The Poynting Theorem gives us insight into energy storage and transfer in the context of electromagnetics

Poynting vector:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2).$$

Total power intercepted by A:

$$P = \int_A \mathbf{S} \cdot \hat{\mathbf{n}} dA \quad (\text{W}).$$



Time-average power density:

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2).$$

Power Density Carried by Plane Wave

For a plane wave with **E** field :

$$\begin{aligned}\tilde{\mathbf{E}}(z) &= \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) \\ &= (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0})e^{-jkz},\end{aligned}$$

the average power density carried by the wave is:

$$\begin{aligned}\mathbf{S}_{\text{av}} &= \hat{\mathbf{z}} \frac{1}{2\eta} (|E_{x0}|^2 + |E_{y0}|^2) \\ &= \hat{\mathbf{z}} \frac{|\tilde{\mathbf{E}}|^2}{2\eta} \quad (\text{W/m}^2),\end{aligned}$$

Plane Wave in Lossy Medium

For a plane wave travelling in a lossy medium:

$$\begin{aligned}\tilde{\mathbf{E}}(z) &= \hat{\mathbf{x}} \tilde{E}_x(z) + \hat{\mathbf{y}} \tilde{E}_y(z) \\ &= (\hat{\mathbf{x}} E_{x0} + \hat{\mathbf{y}} E_{y0}) e^{-\alpha z} e^{-j\beta z}, \\ \tilde{\mathbf{H}}(z) &= \frac{1}{\eta_c} (-\hat{\mathbf{x}} E_{y0} + \hat{\mathbf{y}} E_{x0}) e^{-\alpha z} e^{-j\beta z},\end{aligned}$$

the power density is :

$$\begin{aligned}\mathbf{S}_{\text{av}}(z) &= \frac{1}{2} \Re [\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*] \\ &= \frac{\hat{\mathbf{z}}(|E_{x0}|^2 + |E_{y0}|^2)}{2} e^{-2\alpha z} \Re \left(\frac{1}{\eta_c^*} \right).\end{aligned}$$

$\eta_c = |\eta_c| e^{j\theta_\eta},$

$\mathbf{S}_{\text{av}}(z) = \hat{\mathbf{z}} \frac{|\tilde{E}(0)|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \text{ (W/m}^2\text{)}$

Whereas the fields $\tilde{\mathbf{E}}(z)$ and $\tilde{\mathbf{H}}(z)$ decay with z as $e^{-\alpha z}$, the power density $\mathbf{S}_{\text{av}}$ decreases as $e^{-2\alpha z}$.

Example 7-6: Power Received by a Submarine Antenna

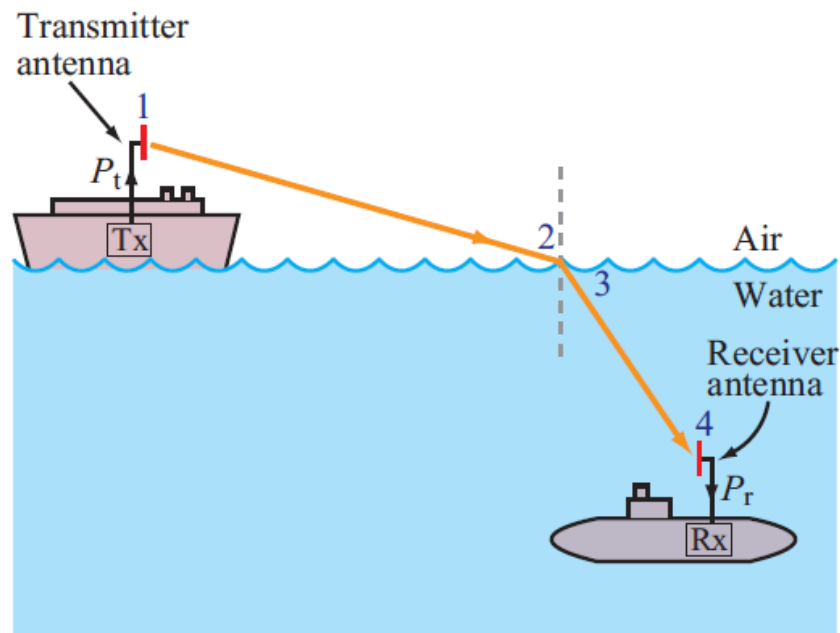
A submarine at a depth of 200 m below the sea surface uses a wire antenna to receive signal transmissions at 1 kHz. Determine the power density incident upon the submarine antenna due to the EM wave of Example 7-4.

Solution: From Example 7-4, $|\vec{E}(0)| = |E_{x0}| = 4.44$ (mV/m), $\alpha = 0.126$ (Np/m), and $\eta_c = 0.044 \angle 45^\circ$ (Ω). Application of Eq. (7.109) gives

$$\begin{aligned} \mathbf{S}_{\text{av}}(z) &= \hat{\mathbf{z}} \frac{|E_0|^2}{2|\eta_c|} e^{-2\alpha z} \cos \theta_\eta \\ &= \hat{\mathbf{z}} \frac{(4.44 \times 10^{-3})^2}{2 \times 0.044} e^{-0.252z} \cos 45^\circ \\ &= \hat{\mathbf{z}} 0.16 e^{-0.252z} \quad (\text{mW/m}^2). \end{aligned}$$

At $z = 200$ m, the incident power density is

$$\begin{aligned} \mathbf{S}_{\text{av}} &= \hat{\mathbf{z}} (0.16 \times 10^{-3} e^{-0.252 \times 200}) \\ &= 2.1 \times 10^{-26} \quad (\text{W/m}^2). \end{aligned}$$



Summary

Chapter 7 Relationships

Complex Permittivity

$$\begin{aligned}\epsilon_c &= \epsilon' - j\epsilon'' \\ \epsilon' &= \epsilon \\ \epsilon'' &= \frac{\sigma}{\omega}\end{aligned}$$

Lossless Medium

$$\begin{aligned}k &= \omega\sqrt{\mu\epsilon} \\ \eta &= \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega) \\ u_p &= \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \quad (\text{m/s}) \\ \lambda &= \frac{2\pi}{k} = \frac{u_p}{f} \quad (\text{m})\end{aligned}$$

Wave Polarization

$$\begin{aligned}\tilde{\mathbf{H}} &= \frac{1}{\eta} \hat{\mathbf{k}} \times \tilde{\mathbf{E}} \\ \tilde{\mathbf{E}} &= -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}\end{aligned}$$

Maxwell's Equations for Time-Harmonic Fields

$$\begin{aligned}\nabla \cdot \tilde{\mathbf{E}} &= 0 \\ \nabla \times \tilde{\mathbf{E}} &= -j\omega\mu\tilde{\mathbf{H}} \\ \nabla \cdot \tilde{\mathbf{H}} &= 0 \\ \nabla \times \tilde{\mathbf{H}} &= j\omega\epsilon_c\tilde{\mathbf{E}}\end{aligned}$$

Lossy Medium

$$\begin{aligned}\alpha &= \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} - 1 \right] \right\}^{1/2} \quad (\text{Np/m}) \\ \beta &= \omega \left\{ \frac{\mu\epsilon'}{2} \left[\sqrt{1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2} + 1 \right] \right\}^{1/2} \quad (\text{rad/m}) \\ \eta_c &= \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}} \left(1 - j\frac{\epsilon''}{\epsilon'} \right)^{-1/2} \quad (\Omega) \\ \delta_s &= \frac{1}{\alpha} \quad (\text{m})\end{aligned}$$

Power Density

$$\mathbf{S}_{\text{av}} = \frac{1}{2} \Re \left[\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^* \right] \quad (\text{W/m}^2)$$

Practice Problems

Problem 7.1 The magnetic field of a wave propagating through a certain nonmagnetic material is given by

$$\mathbf{H} = \hat{\mathbf{z}} 30 \cos(10^8 t - 0.5y) \quad (\text{mA/m})$$

Find the following:

- (a) The direction of wave propagation.
- (b) The phase velocity.
- (c) The wavelength in the material.
- (d) The relative permittivity of the material.
- (e) The electric field phasor.

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- (a) The direction of wave propagation.
- (b) The phase velocity.
- (c) The wavelength in the material.
- (d) The relative permittivity of the material.
- (e) The electric field phasor.

Solution:

- (a) Positive y -direction.
- (b) $\omega = 10^8 \text{ rad/s}$, $k = 0.5 \text{ rad/m}$.

$$u_p = \frac{\omega}{k} = \frac{10^8}{0.5} = 2 \times 10^8 \text{ m/s}.$$

- (c) $\lambda = 2\pi/k = 2\pi/0.5 = 12.6 \text{ m}$.

$$(d) \epsilon_r = \left(\frac{c}{u_p} \right)^2 = \left(\frac{3 \times 10^8}{2 \times 10^8} \right)^2 = 2.25.$$

- (e) From Eq. (7.39b),

$$\begin{aligned} \tilde{\mathbf{E}} &= -\eta \hat{\mathbf{k}} \times \tilde{\mathbf{H}}, \\ \eta &= \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{\epsilon_r}} = \frac{120\pi}{1.5} = 251.33 \quad (\Omega), \\ \hat{\mathbf{k}} &= \hat{\mathbf{y}}, \quad \text{and} \quad \tilde{\mathbf{H}} = \hat{z} 30 e^{-j0.5y} \times 10^{-3} \quad (\text{A/m}). \end{aligned}$$

Hence,

$$\tilde{\mathbf{E}} = -251.33 \hat{\mathbf{y}} \times \hat{z} 30 e^{-j0.5y} \times 10^{-3} = -\hat{\mathbf{x}} 7.54 e^{-j0.5y} \quad (\text{V/m}),$$

and

$$\mathbf{E}(y, t) = \Re(\tilde{\mathbf{E}} e^{j\omega t}) = -\hat{\mathbf{x}} 7.54 \cos(10^8 t - 0.5y) \quad (\text{V/m}).$$

Problem 7.4 The electric field of a plane wave propagating in a nonmagnetic material is given by

$$\mathbf{E} = [\hat{\mathbf{y}} 3 \sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}} 4 \cos(\pi \times 10^7 t - 0.2\pi x)] \quad (\text{V/m})$$

Determine

- (a) The wavelength.
- (b) ϵ_r .
- (c) $\mathbf{H}$.

Problem 7.4 The electric field of a plane wave propagating in a nonmagnetic material is given by

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Determine

- (a) The wavelength.
- (b) ϵ_r .
- (c) $\mathbf{H}$.

Solution:

- (a) Since $k = 0.2\pi$,

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{0.2\pi} = 10 \text{ m}.$$

- (b)

$$u_p = \frac{\omega}{k} = \frac{\pi \times 10^7}{0.2\pi} = 5 \times 10^7 \text{ m/s}.$$

But

$$u_p = \frac{c}{\sqrt{\epsilon_r}}.$$

Hence,

$$\epsilon_r = \left(\frac{c}{u_p} \right)^2 = \left(\frac{3 \times 10^8}{5 \times 10^7} \right)^2 = 36.$$

- (c)

$$\begin{aligned} \mathbf{H} &= \frac{1}{\eta} \hat{\mathbf{k}} \times \mathbf{E} = \frac{1}{\eta} \hat{\mathbf{x}} \times [\hat{\mathbf{y}} 3 \sin(\pi \times 10^7 t - 0.2\pi x) + \hat{\mathbf{z}} 4 \cos(\pi \times 10^7 t - 0.2\pi x)] \\ &= \hat{\mathbf{z}} \frac{3}{\eta} \sin(\pi \times 10^7 t - 0.2\pi x) - \hat{\mathbf{y}} \frac{4}{\eta} \cos(\pi \times 10^7 t - 0.2\pi x) \quad (\text{A/m}), \end{aligned}$$

with

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \simeq \frac{120\pi}{6} = 20\pi = 62.83 \quad (\Omega).$$

Problem 7.6 The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$\mathbf{E} = \hat{\mathbf{y}} 20 \cos(6\pi \times 10^9 t - kz) \quad (\text{V/m})$$

Determine:

- (a) f , u_p , λ , k , and η .
- (b) The magnetic field $\mathbf{H}$.

7.6 The electric field of a plane wave propagating in a lossless, nonmagnetic, dielectric material with $\epsilon_r = 2.56$ is given by

$$\mathbf{E} = \hat{\mathbf{y}} 20 \cos(6\pi \times 10^9 t - kz) \quad (\text{V/m})$$

Determine:

- (a) f , u_p , λ , k , and η .
- (b) The magnetic field $\mathbf{H}$.

Solution:

(a)

$$\omega = 2\pi f = 6\pi \times 10^9 \text{ rad/s,}$$

$$f = 3 \times 10^9 \text{ Hz} = 3 \text{ GHz,}$$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{\sqrt{2.56}} = 1.875 \times 10^8 \text{ m/s,}$$

$$\lambda = \frac{u_p}{f} = \frac{1.875 \times 10^8}{3 \times 10^9} = 6.24 \text{ cm,}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{6.24 \times 10^{-2}} = 100.8 \text{ rad/m,}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{\sqrt{2.56}} = \frac{377}{1.6} = 235.62 \text{ } \Omega.$$

(b)

$$\begin{aligned} \mathbf{H} &= -\hat{\mathbf{x}} \frac{20}{\eta} \cos(6\pi \times 10^9 t - kz) \\ &= -\hat{\mathbf{x}} \frac{20}{235.62} \cos(6\pi \times 10^9 t - 100.8z) \\ &= -\hat{\mathbf{x}} 8.49 \times 10^{-2} \cos(6\pi \times 10^9 t - 100.8z) \quad (\text{A/m}). \end{aligned}$$